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Part I: Reading comprehension: Read the following passage and answer the questions that follow.

Digitization: A Metaphysical Revolution of the Utopia

In 1973, the agronomist René Dumont brought ecological themes starkly to the fore with a book whose title held up an alternative: *L'Utopie ou la Mort* [Utopia or Death]. The readily understandable choice in favor of utopia implied the capacity to imagine another world and a human subject convinced that it was his or her calling to transform it. It expressed humanity's confidence in the resources of its will. Forty years later, the withering of ideals and the general disaffection with the notion of a brighter tomorrow may be such as to require a translation of the "or" in "utopia or death" which would annul the mutual exclusivity of the terms, and thus even the possibility of choosing between them. Latin provides us with the appropriate translation via the passage from the disjunction *aut*, *Utopia aut mortem*, to the coordinating conjunction *vel*, *Utopia, vel mortem* – "utopia, that is, death." In April 2000, the Bibliothèque nationale de France contributed to the conceptual slippage of the expression with an exhibition that retraced the history of the notion of utopia. Throughout, this history moved from the brightest to the darkest of moods, and it ended by conjuring up the totalitarian nightmares of the twentieth century, suggesting that human beings have failed to conceive of an ideal society. Hence, the *idealism* – *virtualism* tension has continued in the digital world as a metaphysical revolution.

What, then, is specific to the utopias generated by today's technological acceleration? The determining factor in the emergence of the new technologies was the development of an apparently very simple procedure: the conversion of all things into a series of 0s and 1s. This procedure, called digitization, is so commonplace today that one would have to be naïve to still be surprised that our world is considered as a collection of discrete elements transformed into signals (whether video, image, audio, printed, genetic, electrical pulse, and so forth) – signals that can be transported, stockpiled, and manipulated.

The result of our contemporary thinking and way of life, digitization is a procedure which homogenizes everything that exists and makes it interchangeable, including of course the very human beings that we are. The digital human being is not only the one who uses today's technological tools; he is also the one to whom these are applied (in the form of scanners, bar codes, genomic sequencing, biometric quantification, electronic prostheses, intracerebral implants, and so on).

Digitization is the crowning triumph of the metaphysical revolution, long the dream of philosophers, as seen behind Descartes's *Mathesis Universalis* or Hegel's *System of Absolute Knowledge*. It makes a general equivalent available (the series of 0s and 1s and the algorithms which process this data), which allows us to unify the whole of reality – and on that basis, to master the world.

It is thus not insignificant that the major utopia of the information society feeds into the processes of globalization – a globalization which began with Christopher Columbus in 1492, and caught a second wind with the industrial revolution, before culminating towards the year 2000 with the creation of instruments that allowed individuals to act at the planetary level. One of the most striking pronouncements on globalization was made by Thomas Friedman, whose 2006 book announced that "the world is flat," Thomas Friedman, *The World Is Flat: A Brief History of*

the... that it shrinks and is flattened at the whim of the ICTs that squeeze time and space, imposing everywhere the same behavior and the same risks (such as those of a stock exchange entrusted to robot traders).

Friedman's book opens with the symbol of the call centers that employ individuals located all over the world. As he writes,

This flat world results from the convergence of the personal computer (which has allowed everyone to produce their own digital documents), fiber-optic cables (which have made it possible to access more and more information at almost no cost), and the development of workflow software (which has allowed individuals to collaborate on the same digital content whatever the geographical distance between them).

It is not idle to stress that individuals constitute the agents of this globalization.

The utopia of a flat world offers comfort to liberal and individualist values; it makes itself acceptable to them in breaking with the specter of mass societies. To take the example of the cultural industries, such as television: what TNT and Connected TV promise is the triumph of the segmentation of the market and the public, such that individualization can still imagine that it is saved from mass mediatization. Drawing on advertising's ceaseless revelation of individualist fantasies, Bauman importantly noted the extent to which the notion of utopia has taken refuge in the comfort of the habitat, in furnishings, and in the journey.

Globalization makes all the elements of the utopia brought about by the digital age widely available: unlimited access to knowledge, peaceful exchange between human beings, the abolition of geographical and ontological borders, collective intelligence, and so on. It is unsurprising then that many forms of spirituality more or less inspired by Buddhism have allied themselves with globalization, declaring that the Internet allows for the fulfillment of the cardinal virtues necessary for karma: the sense of the impermanence of all things (determined by the flux to which we are exposed), the conviction that all things are linked in one great Whole (the interconnection facilitated by cyberspace), and the revelation of the vacuity of the I (the sense of the interchangeability of the neurons that we represent at the level of the planetary brain).

Pierre Lévy, the technologist who famously converted to Buddhism, believes that cyberspace is succeeding in transforming us into "flames" circulating ubiquitously, that is, without constraint. Cyberspace spiritualizes us because it dematerializes us. Such is the mystical argument willingly put forward by the Internet worldview, which sometimes appeals to the notion of the arc leading toward Teilhard de Chardin's Omega Point.

But all technology is, as Plato said, a *pharmakon*: both a poison and a remedy. Technological globalization does not escape this ambiguity. It expresses the utopia of generalized communication (between human beings, between nations, and between objects), but it also creates many opaque spaces, fractures, inequalities, and even forms of insidious violence. In simplifying and rendering lapidary many forms of exchange, it gives rise to behavior that is elementary, uninhibited, unfaithful, brutal, and irresponsible, in a manner that is absolutely at odds with the notion of "tact" cultivated from Goethe to Adorno.

2). The digital world:

- A. Interferes with the religious world
- B. Makes the physical world less dynamic
- C. Destabilizes the comfort zones of many "conservative" people
- D. Transforms the static and discrete world into a dynamic system with multiple constraints.
- E. All of the above

3). In digitization:

- A. The general equivalence is perfectly available to all
- B. The Whole of human reality is unmistakably unified
- C. All things are coming to a pattern of similarity and non-spatiality
- D. The physical volume of the earth shrinks drastically
- E. None of the above

4). Which of the following sentences is correct according to the reading passage?

- A. The flat world is created because of a geophysical phenomena
- B. Workflow softwares are geographical space dependant
- C. Technology and the cyber space put no holes and traces of human errors and dirts
- D. Some religions believe that the cyber space better spiritualizes people
- E. All of the above

5). According to the author of the above passage, it is true that:

- A. Utopia has been defined to the best level of human satisfaction
- B. The conjured totalitarian nightmare of the 20th century ended in a bright tom tomorrow.
- C. Human beings failed to bring a brighter world and convince the ideal society
- D. Digitization as a metaphysical revolution has solved the idealism- virtualism tension of the digital world
- E. None of the above

C. Reference items: What does each of the following words and phrases refer to in the above passage (5pts).

1). it in "... it dematerializes us ..." (Para-10)

2). he in "... he is also the one to whom these are applied..." (Para-3)

3). nontotalitarian universal in "...nontotalitarian universal of the form dreamed..." (Para-12)

4). it in "... it shrinks and is flattened..." (Para-5)

5). This in "... This procedure, called ..." (Para-2)

The utopia of *Homo communicans* is one in which we are desubstantialized and subjected to a nomadism and restlessness that we pathetically associate with freedom – the freedom of eternal tourists who do not remain anywhere. Cyberspace has come to offer up an image of the future which breaks with the values of the sedentary, secure lifestyle of “yesterday’s world.” In the best of cases, it is invoked as the realization of what Pierre Lévy calls an “open universal:” a nontotalitarian universal of the form dreamed about in the American counterculture and by the ’68 generation. In the worst of cases, cyberspace is presented as a collective fantasy in which we find ourselves increasingly reduced to simple and ephemeral avatars who are vulnerable to disconnection, as in Gibson’s novel *The Neuromancer*, which gave the Wachowski brothers the premise for their film, *The Matrix*. (Source Cairn-Int.info)

A. True-False items: Based on the comprehension passage, write True if the statement is correct, and write False if it is incorrect (5pts).

- ____ 1). The internet permits and enhances the impermanence of all things.
- ____ 2). Digitization is an attempt to allow multiple moves, life styles, ideologies and power.
- ____ 3). There is a general fear that the digital world and the cyber space would probably be controlled and manipulated by a few.
- ____ 4). You and I are non-recruited agents and actors of globalization and tech-expansion.
- ____ 5). On the digital human being, various technological tools are not applied at all.
- ____ 6). The utopia of globalization is making the universe borderless and putting it on our thumb.
- ____ 7). The digital revolution results in freedom of exercising virtuous and individualistic liberalism.
- ____ 8). Technology brought to hand multiple manners of immorality, amorality and oddism.
- ____ 9). The digital world emanates from the breaking of discrete, sedentary life and living.
- ____ 10). Technology continues to emerge with sophisticated procedures of conversion.

B. Multiple choice items: Write the letter of your choice that contains the best answer to the following five questions (5pts).

- ____ 1). Why is the metaphysical revolution of utopia a challenge?
 - A. Control is very indefinite
 - B. It breaks cherished values and systems of life
 - C. It increasingly disconnects people and their environments
 - D. The pharmakon is imbalanced
 - E. All of the above

D. Vocabulary items: Match the words/phrases listed under column "A" that are taken from the reading passage with their contextual meanings given under column "B" (use options only once) (5pts).

Column A

- ___ 1) Utopia
- ___ 2) Pharmakon
- ___ 3) Collective fantasy
- ___ 4) Individualist
- ___ 5) Algorithm
- ___ 6) Discrete
- ___ 7) Agents
- ___ 8) Cyber space
- ___ 9) Cardinal virtues
- ___ 10) Ubiquitously

Column B

- A. Formula, set of rules or model
- B. The Net, WWW
- C. Interference
- D. Freethinker, maverick
- E. Forces and actors
- F. Useless and mindless
- G. Separate and various
- H. Omnipresent or universal
- I. The venom and the cure
- J. Geographical space
- K. Everything is perfect
- L. Non-Ethiopia
- M. Prudence or goodness
- N. Shared but unlikely wants/feelings of a pleasant situation or event to happen

E. Identifying main idea of the paragraph in a passage: Read the above passage and fill in the following table accordingly (5pts).

No	Paragraph Number	Main Idea
1	Paragraph ____ 1 ____	_____
2	Paragraph ____	The cyber space achieves its goal of transforming people without any challenge
3	Paragraph ____	Digitization as a utopia of metaphysical revolution has been the long dream of many philosophers
4	Paragraph ____ 9 ____	_____
5	Paragraph ____	Tomorrow, as many fear and consider, is the collective fantasy of the discrete digital human being in the cyber space

Part II: Grammar

A: Choose the best answer after reading each of the following questions carefully (7pts).

1. No sooner _____ the hyena at the gate of the compound than she screamed with fear.

A. had she seen C. she had seen

B. she has seen D. saw

2. The above question (question number 1) means-----

A. She did not scream with fear when she saw the hyena.

B. The lady screamed with fear before she saw the hyena at the gate of the compound.

C. She screamed with fear as soon as she saw the hyena at the gate of the compound.

D. Had she seen a hyena at the gate of the compound, the lady would have screamed with fear.

3. Samrawit: Who gave you my mobile number?

Gugsa: I don't remember. It _____ my friend Tesfa.

Samrawit: It _____ Tesfa. He doesn't know my mobile number.

A. must be/couldn't be

B. must have been/need not have been

C. might have been/mustn't have been

D. may have been/can't have been

4. In the corner of the concert hall _____ a policewoman with a gun, but my friend did not see her at all.

- A. is standing C. stands
B. stood D. has stood

5. I wish I'd visited Meron before she went abroad. This implies that _____

- A. I visited Meron before she went abroad.
B. I visited Meron after she went abroad.
C. I went abroad before I visited Meron.
D. I didn't visit Meron before she went abroad.

6. It was the most interesting discussion that _____

- A. I had ever participated
B. I have ever participated
C. I would have participated
D. I have never participated

7. Kumela: He told me his name was Haile.

Fatuma: You _____ him. His name is Hailu.

- A. should mishear C. must have misheard
B. have to mishear D. can have misheard

B - Dialogue: The following dialogues between Helen and Robel are based on conditional sentences. Depending on the root verbs given in brackets, write logical TENSES and appropriate VERB FORMS (8pts).

Helen: What's wrong, Robel? You look terrible.

Robel: Well, you (1) _____ (look) terrible, too if you (2) _____ (have) a day like mine yesterday. My car slid into a tree because the roads were covered with ice.

Helen: Oh, I was driving on the icy road yesterday, and I didn't have such a trouble. What happened to you?

Robel: Well, I think if I (3) _____ (not drive) so fast, I (4) _____ (not slide) into the tree.

Helen: Icy roads and speed do not mix. If drivers (5) _____ (speed) on ice, they're likely to spin their car in a circle.

Robel: I know, but I've one more problem. I didn't have my driver's license with me. If I (6) _____ (have) it, (7) _____ (not have to) pay an extra fine in the court next month.

Helen: Why were you driving without a license?

Robel: Well, I lost my wallet a few days ago. It slipped out of my pocket while I was riding the bus to work.

Helen: Oh, Robel, if you (8) _____ (not take) that bus, you (9) _____ (not lose) your wallet. If you (10) _____ (not lose) your wallet, you (11) _____ (have) your driver's license with you when you hit a tree. If you (12) _____ (have) your driver's license with you, you (13) _____ (not have to pay) a big fine when you go to court next week. And of course, if you (14) _____ (not drive) too fast, you (15) _____ (not run) into and you (16) _____ (not be) in this mess.

Part III: Paragraph writing

Choose one of the given topics and attempt to write a coherent paragraph (10pts)

(Grammar, punctuation, spelling, capitalization, neatness and a flow of ideas will be considered for marking). (150-200 words).

- ✓ Ethiopian Cultural values that should be promoted.
- ✓ Failure in school does not always predict failure in life.
- ✓ The Contributions of social media for academic purposes.
- ✓ Your reasons to join AASTU.

Part I: True/False Items:

Instruction: Write 'True' if the statement is correct or 'False' if it is incorrect. (1 pt. ea

- F 1. Strength and validity in logic are restricted only to statements and not arguments, and tru
property of arguments but not statements. ✓
- F 2. Fallacies of an informal kind can be identified only with their form and structure.
- T 3. While validity is about inferential claim soundness considers both factual and inferential claims.
- T 4. Attacking a person who presents an argument as opposed to the argument renders an ar
fallacious.
- T 5. Transference of attributes from the whole into the part irrespective of an exception chara
division fallacy.
- T 6. The sufficiency principle synthesizes both acceptability and structural principles.
- F 7. Musical instruments → traditional musical instruments → traditional string musical instrum
Kirar. In this sequence of terms, intension decreases while extension increases. ✓
- F 8. Informal fallacies often occur out of malice on the part of the arguer who presents bad argum
they are good. ✓
- A verbal attack against the reader or the listener is a definitive characteristic of Argum
Misericordiam fallacy.
- F 9. Logical inconsistency means saying one thing and doing another. ✓
- F 10. A verbal attack against the reader or the listener is a definitive characteristic of Argum
Misericordiam fallacy. ✓

Part II: Matching Items

Instruction: Match the items under Column A with those under Column B (1 pt. ea

Column A	Column B
M1. Extensional meaning	A. A defense against serious criticisms
P2. Implicit inferential claim	B. Portrayal of an opponent as hypocrite
P3. Critical thinkers	C. Alleged chain of reaction without sufficient r
F4. Term	D. Resulted from an ambiguity of a key word
A5. Rebuttal principle	E. A standard in critical thinking
P6. Equivocation	F. An object in a statement
B7. Adhominem you too	G. Roughly interpreted as 'sense'
I8. Straw man	H. An argument from an effective and exhaust
C9. Slippery Slope	I. Distorting opponents' argument for attacki
N10. Hasty generalization	J. AASTU regular students
	K. Practice restraint
	L. Resulted from an ambiguity in sentence s
	M. Roughly interpreted as 'reference'
	N. Argument from a premature sampling
	O. Reasoning process without indicator wor
	P. Preoccupied with themselves and their o
	Q. Tracing circumstance of an opponent fo

Part III: Multiple-choice Items (1 pt. each)

Instruction: Choose the correct answer from the given alternatives.

1. "You argue that I should cut down on my drinking. But you drink much more heavily than I. You haven't been sober in a year." The fallacy committed in this argument:
- ☒ A. Ad hominem, you too B. Unqualified authority C. Slippery slope D. Missing the point E. Ad hominem, circumstantial
2. One of the following is fallacious because of its form or structure:
- A. Surely Rebel loves me. For he told me he loves me, and he wouldn't lie to someone he loves.
 B. After Hanna returned from work the other day, she switched on her air conditioner, and a few seconds later a power failure struck the city. Therefore, to prevent further power failures, Hanna should keep her air conditioner shut off.
 C. Jamal says that we shouldn't have a central bank because central banking is responsible for the economic fluctuations of the business cycle. But people have been banking for centuries. Bankers aren't bad people, and they provide the valuable service of providing credit to people who don't have their own capital.
 D. If Russia invades Ukraine, then Russia wants war. Russia wants war. So, Russia invades Ukraine.
 E. If one were to listen only to one kind of music or eat only one kind of food, it would soon become tasteless or boring. Variety makes eating and listening exciting and enriching experience. In a similar manner, an exclusive sexual relationship with only one partner for the rest of one's life, that is, marriage, would soon become tasteless or boring. Thus, a plenty of extra-marital sex makes life exciting and enriching.
3. A deductive argument with all true premises and false conclusion is always:
- A. Strong B. Valid C. Sound ☒ D. Invalid E. Weak
4. The definition "integration" means a communist "inspired policy that forces rich parents to send their children to inferior schools" is an example of _____ definition.
- A. Theoretical ☒ B. Persuasive C. Lexical D. Précising E. Operational
5. "Let's use 'bromance' to mean a movie about the relationship of two men who have a very close friendship." The definition is:
- A. Lexical B. Persuasive C. Theoretical ☒ D. Stipulative E. Précising
6. In one of the following forms of argument, the content of the conclusion is in some way intended to "go beyond" the content of the premises.
- A. An argument that depends on the existence of an analogy, or similarity, between two things or states of affairs.
 B. An argument that proceeds from our knowledge of the past to a claim about the future.
 C. An argument in which the conclusion depends on some purely arithmetic or geometric computation
 D. All except 'C' E. None

B 7. Which of the following consists of the properties or attributes that the term commonly connotes to the member of the community represented by the term?

- A. Empty extension
- B. Extensional meaning
- C. Conventional connotation
- D. Intentional meaning
- ☒ E. C and D

E 8. When each term in a given series connotes less attributes than the one preceding it, then the series is said to have been typically arranged in the order of:

- A. Decreasing extension
- B. Decreasing intension
- C. Increasing extension
- D. Increasing intension
- E. none

B 9. "The number of alcoholics has risen steadily for the past 20 years, and so has the number of treatment centers. Therefore, to cut down on alcoholism we should eliminate the treatment centers." The argument commits the fallacy of:

- ☒ A. Missing the point
- B. Non-cause pro-cause
- C. Appeal to people
- D. Accident

D 10. "Last year, the Software Engineering student Abdissa used blue Lexi pen to solve the problems on logic final exam, and after the exam was graded he got an 'A'. So if you want to get an 'A' in logic, you should work the entire exam question with a blue Lexi pen". The argument commits the fallacy of:

- A. Adhominem-you too
- B. False Dichotomy
- ☒ C. False cause
- D. Begging the question

B 11. "Telepathy does not exist since nobody has shown any prove that it is real". This argument commits the informal fallacy of:

- ☒ A. Appeal to ignorance
- B. Appeal to arrogance
- C. Appeal to authority
- D. Appeal to vanity

B 12. One of the following is an example of a **Precising** definition:

- A. "Lawyer" means dishonest and greedy practitioner of the law who will say anything to win a court case and make money.
- B. "Too hot" for purposes of judging the temperature of coffee means hotter than 180 degrees Fahrenheit.
- C. "Poncho" means rain jacket.
- D. "Boiling" means bubbles of saturated vapor forming in the liquid phase of matter.

C 13. Critical thinkers exhibit the following characters except one.

- A. They are interested in other people's ideas
- B. They are not controlled by their feelings
- ☒ C. They pretend they know more than they actually do
- D. They recognize their limitations

14. Informal fallacies are committed because:
- The premises are logically irrelevant to establish their conclusion.
 - The premises are not sufficient enough to accept their conclusion.
 - Some premises use ambiguous terms in their premise/conclusion.
 - All
15. Which one of the following fallacies is categorized under the fallacies of Presumption?
- Post hoc ergo propter hoc
 - False dichotomy
 - False cause
 - Slippery slope
 - All
16. "Emperor Tewodros II is the grand architect of Ethiopian modernity. He tried to revolutionize existing relations between the state and church; he tried to build a modern army and also sought to establish foreign relations with western European nations." The passage is:
- Argument; conclusion: He tried to revolutionize existing relations between the state and church.
 - Argument; conclusion: he tried to build a modern army.
 - Expository passage
 - Loosely associated statements
 - Argument; conclusion: Emperor Tewodros II is the grand architect of Ethiopian modernity.
 - All
17. "Yesterday I bought new jeans with a tag that reads, 'Made in U.S.A.'. Thus, my new jeans were made in the U.S.A." The argument is
- Deductive, Uncogent
 - Inductive, Strong
 - Deductive, Unsound
 - Deductive, valid
 - Inductive, Weak
18. "All managers are journalists. All journalists are lawyers. So, all lawyers are journalists". This argument is said to be:
- Uncogent Syllogism
 - Invalid Argument
 - Valid Categorical syllogism
 - Cogent Syllogism
 - Valid Hypothetical Syllogism
 - Invalid Disjunctive Syllogism
19. "No cat is reptiles owing to all cats are dogs and no dog is a reptile." What would be the proper name of this argument in terms of both inferential and factual claims?
- Sound
 - Unsound
 - Valid
 - Invalid
 - C and B
 - D and B
20. Distinguishing between deductive and inductive arguments may sometimes be difficult. It generally requires an effective assessment of the strength or otherwise of their inferential claims; the strength of the inferential claim of an argument can be determined using one or more of the following mechanisms. Identify it.

- The Form of argumentation
- The actual strength of the inferential claim
- The truth or falsity of the premises
- A and B
- A, B and C

D 21. The argument, "The Ethiopian Intelligence Agency failed to prevent the TPLF attack on members of the Ethiopian National Defense Force. Therefore, this government agency should be abolished immediately," commits;

- A. Complex question. C. Begging the question.
B. No fallacy. ~~D. Missing the point.~~ E. Appeal to ignorance

22. One is **wrong** about uncritical thinkers

- A. Tend to follow their feelings and act impulsively
B. Base judgments on first impressions and gut reactions
~~C. Are patient with complexity.~~
D. Pretend they know more than they do
E. None

23. A **principle** in critical thinking that requires each participant in a discussion of a disputed issue should be willing to accept the fact that he or she is imperfect is

- A. Truth Seeking Principle B. Fallibility Principle C. Clarity Principle D. Charity Principle

B 24. From the following arguments, identify the one susceptible of committing the fallacy of **accident**;

- A. Benito Mussolini was an insincere Italian leader. Therefore all Italians are insincere.
~~B. The Bible says "Thou shalt not kill"; therefore, you should not try to control termites in your home or fight for your country.~~
C. We can ignore what Karl Marx said about capitalism. After all, he was hopeless that he didn't even support his family.
D. You must believe with me that this individual is guilty of murder, for if you do not find him guilty, upon his release, you will probably be his next victim.
E. "No one would blame a bartender for having a few drinks on the job. But a professor in a university is no less a human being than a bartender. So, no one should blame a professor for having a few drinks on the job."

A 25. "Professor Andrew, who is a famous astrophysicist working for NASA, was heard saying, Africa's development has retarded due to bad economic policies.' On the account of Professor Andrew's speech, African countries are following wrong economic policies." The fallacy committed is:

- A. Appeal to unqualified authority C. Appeal to ignorance
B. Hasty generalization D. Slippery slope E. Missing the point

PART I. TRUE FALSE ITEMS

DIRECTIONS: Write 'True' if the statement is correct and otherwise 'False'
(1.25 points each)

1. Both negative reinforcement and negative punishment decrease the likelihood of a given behavior.
2. Gestalt psychology emphasis on units that make up the mind or human behavior.
3. The degree of retinal disparity is larger when the object is closer to the observer.
4. Whether our sensory input is figure or ground it depends on our attention.
5. Punishment can be applied for the purpose of increasing aversive behavior.
6. People with panic attack can predict when and where the panic attack will occur.
7. As the memory is the vital aspect of life, forgetting has no contribution except disrupting once life.
8. Elaborative rehearsal is mostly effective to hold information in short term memory as a focus of attention.
9. According to humanistic theory, the ultimate goal of personality growth is self-actualization.
10. Schechter Singer theory considered emotions occur as a result of physiological reactions to events.

PART II: MULTIPLE CHOICE ITEMS

DIRECTIONS: For each of the following items, choose the best answer from the given four alternatives and write the letter of your choice in the space provided on the answer sheet (1.25 points each)

11. Mr. X works in a mobile maintenance shop and produces 10 mobiles every day on average. He earns money after assembling 150 mobiles. What is the kind of reinforcement that Mr. X received?
 - A. Fixed interval
 - B. Fixed ratio
 - C. Variable ratio
 - D. Variable interval
12. When information arising from real external stimuli leads to an incorrect perception it is known as:
 - A. Depth perception
 - B. Perceptual illusion
 - C. Perceptual consistency
 - D. Perceptual organization
13. Positive punishment _____ the behavior and negative reinforcement _____ the behavior.
 - A. Strengthens, Weakens
 - B. Strengthens, Strengthens
 - C. Weakens, Strengthens
 - D. Weakens, Weakens

14. The ego defense mechanism in which a person who is confronted with a more immature behavioral stage is called _____
- A. Repression
 - B. Regression
 - C. Reaction formation
 - D. Rationalization
15. Which of the following is related with cognitive view of psychological disorder?
- A. Behavior is the result of conflict of intra-psychic forces.
 - B. Normality in behavior depends on adaptive thinking.
 - C. Normality in behavior is the result of learning from the environment.
 - D. Behavior is influenced by physiological problem in central nervous system.
16. Suppose the ego were a hungry person is in a crowded cafeteria. The ego would most likely
- A. Do nothing.
 - B. Feel guilty about being so hungry
 - C. Wait in line
 - D. Push to the head of the line.

Case Given: Read the case scenario below and answer question 17 and 18.

Roza's dorm is not clean even she is not reluctant to clean. But, now her dorm mates agree to pay her an allowance of 20 birr per week if she cleans her dorm every Saturday and they told her they put additional dust which is a cause for her asthma if she does not agree on allowance.

17. The first method that Roza's dorm mates used to clean the dorm applied which technique to modify her behavior?
- A. Positive reinforcement
 - B. Negative reinforcement
 - C. Positive punishment
 - D. Negative punishment
18. The second method that Roza's dorm mates used is _____
- A. Positive reinforcement
 - B. Negative reinforcement
 - C. Positive punishment
 - D. Negative punishment
19. The smallest difference between two stimuli that can be detected 50 percent of the time it is present is called _____
- A. Absolute threshold
 - B. Sensation
 - C. Just noticeable difference
 - D. Sensory adaptation
20. Smoke detectors, preventive medicines, insurance, retirement investments, seat belts, burglar alarms, and sunscreen are all examples of products to satisfy consumers' _____ needs.
- A. Safety
 - B. Self-actualization
 - C. Belongingness
 - D. Esteem
21. Which of the following pairs accurately describes the two central characteristics of Short-Term Memory?
- A. Limited time; unlimited capacity
 - B. Limited time; limited capacity
 - C. Unlimited time; limited capacity

D. Unlimited time; unlimited capacity

22. Student X is currently attending his education in AASTU. He is hard working, enthusiastic to learn up-to-date scientific principles. However, he wants to be alone most of the time in the in the University. Student X's personality is highly related with _____

- A. High neuroticism and less for openness to experience,
- B. Low extraversion and high for openness to experience
- C. High for both conscientiousness and openness to experience,
- D. High for agreeableness and low for extraversion

23. Referring the picture below, just consider yourself standing in the middle of a cobblestone street, the street will look *convex* near your feet and *linear* if you look into the distance. This is called:



- A. Texture gradient
- B. Relative size
- C. Motion parallax
- D. Linear perspective

24. While hearing a sequence of sounds of different pitches it is to _____ recognizing the sound sequence as a familiar melody is to _____

- A. Absolute threshold; difference threshold
- B. Signal detection; sensory adaptation
- C. Sensory adaptation; signal detection
- D. Sensation; perception

25. The tendency of a person to group close things together can be referred as:

- A. Similarity
- B. Closure
- C. Continuity
- D. Proximity

26. One of the following research methods is often concerned with detail study of particular person

- A. Survey
- B. Experimental method
- C. Naturalistic observation
- D. Case study

27. We can quantify friendliness and kindness and determine the degree to which they apply to a specific person. Which theory of personality supports this statement?

- A. Psychoanalytic theory
- B. Trait theory
- C. Humanistic theory
- D. Learning Theory

PART II: MATCHING ITEMS

DIRECTION: Match Concepts and Theories listed under Column "A" with their best Description or characteristic under "B" (1 point each)

- | <u>A</u> | <u>B</u> |
|---------------------------------------|--|
| 28. Drive-Reduction Theory | A. Characterized by excessive attention seeking. |
| 29. Narcissistic personality disorder | B. Liking an activity or object to obtain reinforcement |
| 30. Borderline personality disorder | C. Facial movements elicit specific emotions |
| 31. Histrionic personality disorder | D. Exaggerated self-worth, and need for admiration |
| 32. James-Lange Theory | E. Sustain and directs behavior |
| 33. Facial Feedback Hypothesis | F. Feeling responses to thoughts, situations, or behaviors |
| 34. Motivation | G. Motivation resulting from liking an action or object itself |
| 35. Emotions | H. Feelings interpreted from physical sensations |
| 36. Intrinsic Motivation | I. Internal tensions "push" organism toward basic needs |
| 37. Extrinsic Motivation | J. Bouts of clinginess in personal relationships |

PART II: FILL IN THE BLANK SPACE ITEMS

DIRECTION: For questions 38-40, fill in the blank space with the appropriate concept, theory, or perspective and put your answer on the answersheet (1.25 points each)

38. Among schools of psychology, _____ perspective emphasizes the unconscious dynamics within the individual such as inner forces, unfulfilled wishes and desires, conflicts or instinctual energy.
39. A defense mechanism that the individual may engage in by transforming psychic energy from aggressive behaviors and sexual urges to adaptive and acceptable behaviors is _____.
40. _____ theory proposes that people are different in their capacity to detect the presence of stimulus. As a result, sensitivity to environmental stimulus is different which will lead to different decisions for similar stimulus/information presented.

Part I Write the most simplified form of your answers only on the space provided for the following items. Each item will be awarded 2 marks for complete answers.

1. Let p and q be propositions which stands for

$$p(x) : |x| = -1 \quad q(x) : e^{2x} - 4 = 0$$

then the truth value of $(\exists x)(p(x) \Rightarrow q(x))$ is True (T) [02]

2. Let $A \cup B = \{a, b, d, e, x, y, z\}$ and $A \cap B = \{b, e, y\}$.

(a) If $B - A = \{x, z\}$, then $A = \{b, e, y, a, d\}$

(b) If $A - B = \emptyset$, then $B = A \cup B = \{a, b, d, e, x, y, z\}$

[02]

3. For propositions p, q and r , let the truth value of the compound proposition $\neg[\neg r \Rightarrow \neg(p \wedge q)]$ be True, then the truth value of

(a) $(p \wedge q) \Rightarrow (\neg q \vee p)$ is True (T)

(b) $\neg(p \Leftrightarrow (q \vee r))$ is False (F)

4. If $z = \sqrt{2}(\cos(\frac{\pi}{24}) + i\sin(\frac{\pi}{24}))$, then the value of z^4 is $2\sqrt{3} + i$ [02]
 $4[\cos(\frac{\pi}{6}) + i\sin(\frac{\pi}{6})] = 4e^{i\frac{\pi}{6}}$ [02]

5. The lub and glb of the set $S = \{3.01, 3.0101, 3.010101, \dots\}$ are $3.\overline{01}$ and 3.01 [02]

6. The inverse of the function $f(x) = \frac{x+1}{x-1}$, $x \neq 1$ is $f^{-1}(x) = \frac{x+1}{x-1}$ [02]

7. Let $f(x) = x^2 - 3x$ and $g(x) = 1 + x$, then the solution of $f(g(x)) = 0$ is $\{-1, 2\}$ [02]

8. For what value of k , $x^2 + y^2 - 6y + k = 0$ represents a circle? $k \leq 9$ [02]

9. The vertex, focus and directrix of the parabola $x^2 - 8y - 6x + 1 = 0$ are $(3, -1)$, $(3, 1)$ and $Y = -3$ respectively. [02]

10. The eccentricity of the ellipse $\frac{(x-2)^2}{4} + \frac{(y+1)^2}{9} = 1$ is $\frac{\sqrt{5}}{3}$ [02]

[02]

Part II Show all the necessary steps of all questions in this section using the answer space provided.

11. Show that $\sqrt{2}$ is an irrational number.

[03]

proof: suppose that $\sqrt{2}$ is rational number.

$\Rightarrow \sqrt{2} = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ and $\text{G.C.F}(a, b) = 1$

$$\Rightarrow 2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$$

$$\Rightarrow a^2 \text{ is even}$$

$$\Rightarrow a \text{ is even}$$

$$\Rightarrow a = 2k, k \in \mathbb{Z}$$

$$\Rightarrow 4k^2 = 2b^2$$

$$\Rightarrow b^2 = 2k^2$$

$$\Rightarrow b^2 \text{ is even}$$

$$\Rightarrow b \text{ is even}$$

$$\Rightarrow \text{GCF}(a, b) \geq 2$$

i.e. a & b are not relatively prime or $\frac{a}{b}$ is not in simplest form.

$$\Rightarrow \text{Which is a contradiction}$$

$$\Rightarrow \sqrt{2} \text{ is not rational no.}$$

Hence, $\sqrt{2}$ is an irrational no.

12. Find the fourth root of $-1 + \sqrt{3}i$.

[04]

$$\text{Let } Z = -1 + \sqrt{3}i$$

$$\Rightarrow r = |Z| = 2$$

$$\Rightarrow \theta = \text{Arg}(Z) = \tan^{-1}(\sqrt{3})$$

$$\therefore \theta = \frac{5\pi}{6}$$

$$\Rightarrow \sqrt[n]{Z} = r^{1/n} e^{i(\frac{\theta + 2k\pi}{n})}$$

$$= \sqrt[n]{2} e^{i(\frac{5\pi/6 + 2k\pi}{4})}$$

$$, n = 4, k = 0, 1, 2, 3$$

$$, k = 0, 1, 2, 3$$

$$= \sqrt[n]{2} \left[\cos\left(\frac{5\pi/6 + 2k\pi}{4}\right) + i \sin\left(\frac{5\pi/6 + 2k\pi}{4}\right) \right]$$

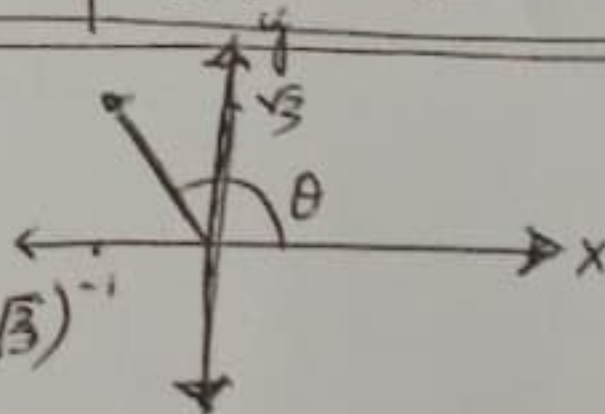
$$Z_0 = \sqrt[n]{2} \left[\cos\left(\frac{5\pi}{24}\right) + i \sin\left(\frac{5\pi}{24}\right) \right]$$

$$Z_1 = \sqrt[n]{2} \left[\cos\left(\frac{17\pi}{24}\right) + i \sin\left(\frac{17\pi}{24}\right) \right]$$

$$Z_2 = \sqrt[n]{2} \left[\cos\left(\frac{29\pi}{24}\right) + i \sin\left(\frac{29\pi}{24}\right) \right]$$

$$Z_3 = \sqrt[n]{2} \left[\cos\left(\frac{41\pi}{24}\right) + i \sin\left(\frac{41\pi}{24}\right) \right]$$

Please go on to the next page



13. Find the value of x for which the graphs of $f(x) = 3^x$ and $g(x) = 4^{-x+2}$ intersect.
[Hint: Use common logarithm]

[05]

Soln. $3^x = 4^{-x+2}$ gives us the point of intersection — (1pt)

$$\Rightarrow \log_{10} 3^x = \log_{10} 4^{-x+2} \quad \text{— } \left. \begin{array}{l} 1 \text{ pt} \end{array} \right\}$$

$$\Rightarrow x \log_{10} 3 = (-x+2) \log_{10} 4$$

$$\Rightarrow x \log_{10} 3 + x \log_{10} 4 = 2 \log_{10} 4$$

$$\Rightarrow x \log_{10} 12 = \log_{10} 16$$

$$\Rightarrow x = \frac{\log_{10} 16}{\log_{10} 12} = \log_{12} 16$$

3 pts

Or

$$3^x = 4^{-x+2} \Rightarrow 3^x = \frac{16}{4^x}$$

$$\Rightarrow 12^x = 16$$

$$\Rightarrow \log_{12} 16 = x$$

14. Suppose the polynomial function $f(x) = 2x^3 + px^2 - 3x + k$. If f is divided by $x + 2$ and $x - 1$, then the remainders are -20 and -2 respectively.

(i) Find the values of p and k .

(ii) Find the zeros of f .

Soln : By remainder theorem, we have

$$\begin{aligned} (i) \quad f(-2) = -20 &\Rightarrow k + 4p = -10 \quad \dots (1) \\ f(1) = -2 &\Rightarrow k + p = -1 \quad \dots (2) \end{aligned}$$

$$\Rightarrow k = 2 \text{ and } p = -3$$

$$\text{Hence, } f(x) = 2x^3 - 3x^2 - 3x + 2$$

(ii) By rational root theorem, we have a zero of polynomial may be one of $\pm 1, \pm \frac{1}{2}, \pm 2$

$$\text{Now, } f(1) = -2 \neq 0$$

$$f(-1) = 0$$

$\therefore x + 1$ is factor of $p(x)$

— And using long division, we get that

$$f(x) = 2x^3 - 3x^2 - 3x + 2 = (x + 1)(2x^2 - 5x + 2)$$

other zero of $f(x)$ will be

$$2x^2 - 5x + 2 = 0$$

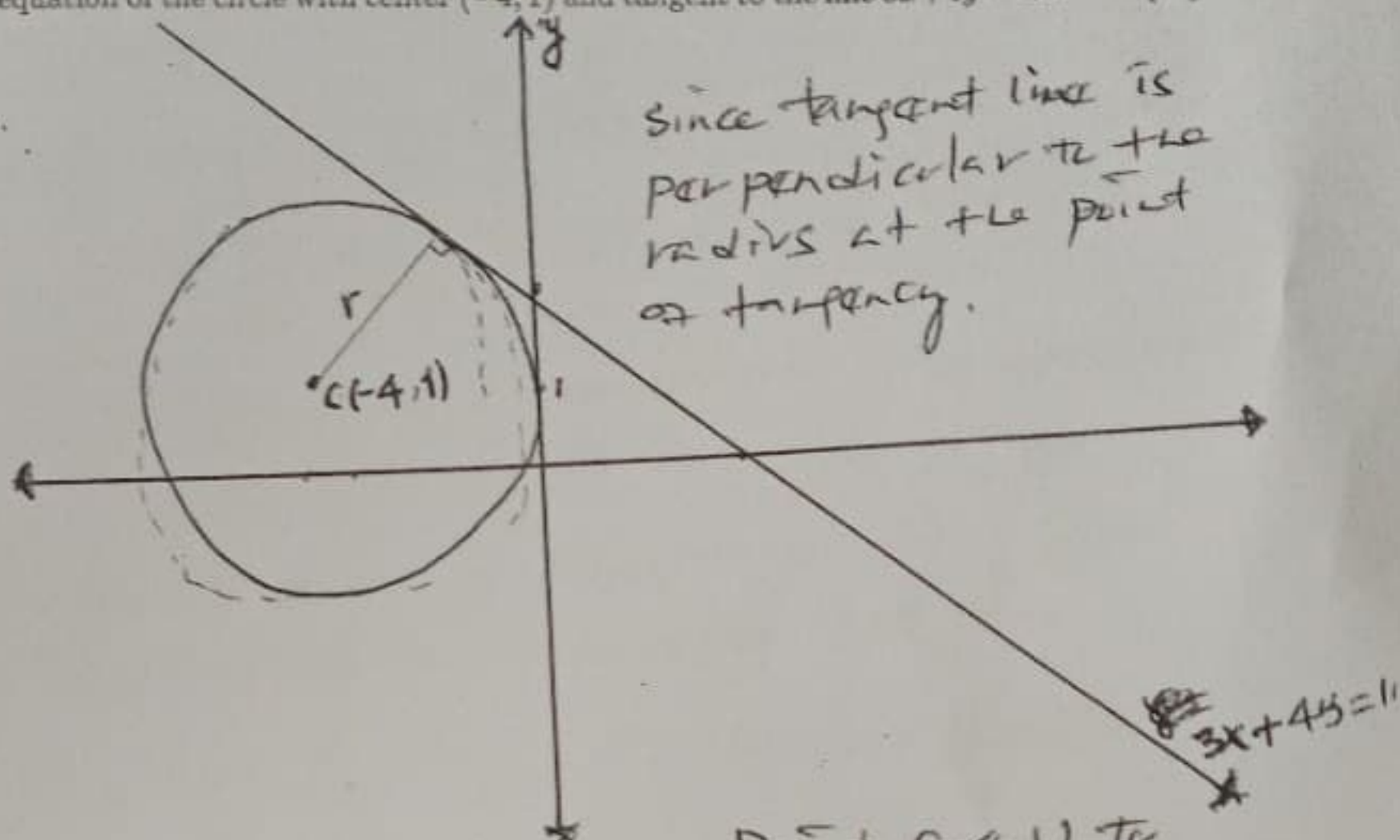
$$x = \frac{+5 \pm \sqrt{25 - 16}}{4} = \frac{+5 \pm \sqrt{9}}{4}$$

$$x = \frac{+5 \pm 3}{4}, \quad x = 2 \text{ and } x = \frac{1}{2}$$

Hence, the zeros of $f(x)$ are $-1, \frac{1}{2}$ and 2 .

15. Find the equation of the circle with center $(-4, 1)$ and tangent to the line $3x + 4y = 10$. [05]

Solⁿ :



Since tangent line is perpendicular to the radius at the point of tangency.

By distance formula from a point $(-4, 1)$ to the line $3x + 4y - 10 = 0$ is the radius of the circle.

$$3pt \left\{ \begin{aligned} r &= \frac{|3(-4) + 4(1) - 10|}{\sqrt{9 + 16}} = \frac{|-18|}{5} = 18/5 \end{aligned} \right.$$

Hence, the equation of a circle will be

$$(x-h)^2 + (y-k)^2 = r^2 \quad \left\{ \begin{aligned} (1pt) \end{aligned} \right.$$

$$\underline{1pt} \left\{ \therefore (x+4)^2 + (y-1)^2 = (18/5)^2 \right.$$

16. Let the length of the transverse of a hyperbola is 16 and the end points of the conjugate axis are $(5, -5)$ and $(5, 3)$, then find:

- The equation of the hyperbola
- The vertex and the foci of the hyperbola.
- Sketch the graph

So (i) eqn of the hyperbola is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

1 Pt

[07]

Where the center $C(h, k) = \text{midpoint of conjugate axis}$

$$\Rightarrow C(h, k) = (5, -1)$$

$a = 8$ (half of length of transverse axis)

$b = 4$ (half of length of conjugate axis)

2 Pt

$$\Rightarrow \frac{(x-5)^2}{64} - \frac{(y+1)^2}{16} = 1 \quad \text{is eqn of the hyperbola.}$$

ii) the vertices are

$$V(h-a, k) \text{ and } V'(h+a, k)$$

$$= V(-3, -1) \text{ and } V'(13, -1)$$

1 Pt

$$c = \sqrt{a^2 + b^2} = \sqrt{80}$$

$\Rightarrow$ the foci are $F(h-c, k) = F(5-\sqrt{80}, -1)$ and

$$F'(h+c, k) = F'(5+\sqrt{80}, -1)$$

1 Pt

